

Isabella A. Stewart

istewart@mit.edu

EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT)

Ph.D. in Civil and Environmental Engineering; *Mechanics and Materials* (Grade: 5.0/5.0)

Cambridge, MA

SEPT 2024 - Present

WELLESLEY COLLEGE

B.A. in Chemistry with Honors, Sigma - Xi Honor Society

Wellesley, MA

SEPT 2019- DEC 2023

AWARDS

- Department of Energy (DOE) Computational Science Graduate Fellowship — Awarded (April 2025)
- Whitaker Health Sciences Fund Fellowship — Awarded (April 2025)
- National Science Foundation (NSF) Graduate Research Fellowship — Awarded (April 2024), **declined in favor of DOE CSGF**
- MIT GSO Gifts Fellowship (9 month department fellowship) — Awarded (March 2024)

RESEARCH EXPERIENCE

THE AHMED LAB AT MIT (DeCoDE LAB)

Under the supervision of Prof. Faez Ahmed, my research focuses on optimization algorithms for high performance structures and materials. I construct multiagent AI architectures endowed with topology optimization (TO) physics solvers to meet human-centered design needs.

Second-Year Graduate Student Researcher

SEPT 2025-Present

- Development of multimodal large language model agents for steering topology optimization landscapes to achieve long-horizon design objectives aligned with human needs and preferences.

THE BUEHLER LAB AT MIT (LABORATORY OF ATOMISTIC AND MOLECULAR MECHANICS)

Under the supervision of Prof. Markus J. Buehler, my research focused on the study of high performance materials with a focus on bioinspired technology. I use multiscale modeling predictions for training machine learning models to generate novel materials capable of withstanding extreme and dynamic environments.

First-Year Graduate Student Researcher

SEPT 2024-SEPT 2025

- Augmenting multi-agent scientific reasoning with large-scale knowledge graphs to generate novel hypotheses for biocomposite materials. **{Two first-author manuscripts}**
- Multiscale simulation methods interfacing with machine learning and additive manufacturing techniques.

Undergraduate Student Thesis Researcher

SEPT 2022- AUG 2024

- Proposed, developed, and performed molecular dynamic simulations to biocoat titanium dioxide surfaces with collagen. Used simulation trajectories to design implant coatings for osseointegration of titanium implants with the human body. Undergraduate thesis: "Enhanced Osseointegration of Metal Oxide Implants: A Computational Study" **{First-author published thesis}**
- Developed tools and scripts to improve the efficiency of the interfacial modeling of the abio-bio interface. Optimized high process computing using enhanced sampling techniques.

THE NAZARIAN LAB AT HARVARD MEDICAL SCHOOL (BIDMC)

Researcher

JUN 2021- AUG 2024

Under the supervision of Dr. Ara Nazarian, I performed computational studies at the continuum level by finite element analysis (FEA) and experimentally characterized bone and tissue under large deformation and stresses. Over the course of three years I was involved in a variety of projects to address orthopedic challenges as follows:

- Novel Bending Mechanical Test for Bone Studies: Identified inconsistent reports of mechanical indices due to three and four point bend testing methodology for mouse femora in literature, proved faults in the test frame with a finite element model, and designed a new test frame and method called 'Contactless Bending' method to establish uniform bend testing. **{Two first-author manuscripts accepted to Journal of Biomechanical Engineering}**
- FEA of the Rotator Cuff: Collaborated in building a finite element model for a rotator cuff tear study, performed sensitivity test analysis, and completed data acquisition in MATLAB **{Published manuscript}**
- Stemless Arthroscopy Implant: Built a mechanical testing system to investigate strain response in cadaveric humerus with a shoulder implant: built DAQ circuitry, coded data acquisition and analysis in MATLAB and Arduino **{Published manuscript}**
- Etiology of Tendinopathy with Rat Tendon Fixture: Investigated tendon mechanical load testing of rat ankle dorsiflexion. Worked on MATLAB code to actuate the testing system and performed data analysis. **{Published manuscript}**
- Shoulder Balloon Stent Implant: Designed a permanent shoulder implant for the subacromial space and investigated stress conditions using a finite element model. Applied a python-based addon to Abaqus to perform balloon shape design. **{Manuscript in preparation}**

THE KAPLAN LAB AT TUFTS UNIVERSITY

Researcher

JAN. 2024-AUG 2024

At the Kaplan Lab, I developed and performed multiscale simulations in combination with material characterization experiments to improve biopolymer mechanics in hydrated environments.

- Thermoplastic silk composites lose mechanical strength in hydrated conditions due to plasticization by the water. As a countermeasure, I studied various mineral and organic long-rod fillers to develop interconnecting rod-networks within a thermoplastic silks composite.
- Developed method for homogeneously distributing the filler in the composite by mixed sonication, freeze-drying, and silk processing techniques. Applied material characterization techniques (tensile testing, point-bending tests, TGA analysis, FTIR, contact-angle studies, water-absorption tests)
- Created and analyzed coarse-grain model of the filler and atomistic model of the filler-silk interface as a high throughput assessment of the experimental results. **{Manuscript in preparation}**

LEADERSHIP

PROJECT W

Co-Founder

JUN 2020 - DEC 2023

Project W is a community of 500+ college women innovators and engineers . We are dedicated to fostering interdisciplinary collaboration amongst women to start high-level projects worthy of investment in industry.

- Delivery of hands-on workshops, startup resources, and industry expertise
- Mentored students in STEM research-based innovations and startups.

WOMEN IN COMPUTATIONAL CHEMISTRY SYMPOSIUM

Lead Organizer

NOV 2024

- Recruited faculty and students to participate in demonstrating tools and to discuss strategies for building an academic trajectory in computational chemistry
- 60+ attendees (all women students at Wellesley College)

WELLESLEY COLLEGE ALUMNAE ASSOCIATION

Elected Co-President

2023- PRESENT

- Organize and foster alumni connection and resources for career including awards and recognitions

SILK INTEGRATIVE THEORY EXPERIMENT PROJECT (SITEP)

Lead Organizer

2024 - PRESENT

- Organize Tufts (David Kaplan Group) and MIT (Buehler Group) Collaboration on USDA projects and Army Research Projects

PUBLICATIONS

- **Stewart, I.**, Buehler, M.J., Higher-Order Knowledge Representations for Agentic Scientific Reasoning. *arXiv preprint arXiv:2601.04878* (2026). [Under Review at Nature Artificial Intelligence]
- **Stewart, I.**, Hage, T.P, Hsu, Y.C., Buehler, M.J., GraphAgents: Knowledge Graph-Guided Agentic AI For Cross-Domain Materials Design [Submission in progress to Nature Computational Materials]
- **Stewart, I.**, Buehler, M.J., Molecular analysis and design using generative artificial intelligence via multi-agent modeling, *Mol. Syst. Des. Eng.*, 2024, Royal Society of Chemistry (RSC).
- **Stewart, I.**, Garcia, M., Hedayatzadeh Razavi, A., Nafisi, N., Buzo, M., & Nazarian, A. (2024) Reducing variability in bone mechanical testing: A contactless bending approach for improved reproducibility. *J Biomech Eng.* 2025 Nov 1;147(11):111006. doi: 10.1115/1.4069500. PMID: 40847595.
- **Stewart, I.**, Garcia, M. J., Alluri, N., Buzo, M., Keko, M., & Nazarian, A. (2024). A meta-analysis study to define variations in murine long bone biomechanical testing. *J Biomech Eng.* 2025 Jun 1;147(6):060801. doi: 10.1115/1.4068318. PMID: 40172045.
- Hedayatzadeh Razavi, A., Nafisi, N., **Stewart, I.**, Abbasian, M., Kheir, N., Shariyate, M. J., Khak, M., Momenzadeh, K., Asciutto, D., Ramappa, A. J., Ross, G., Shah, S., & Nazarian, A. (2024). The biomechanical assessment of two stemless shoulder arthroplasty prostheses in uniformly poor-quality bone mineral density cadaveric specimens. *Clinical Biomechanics*, 120, 106346. DOI: <https://doi.org/10.1016/j.clinbiomech.2024.106346>
- Williamson, P., Garcia, M., Momenzadeh, K., Abbasian, M., Kheir, N., **Stewart, I.**, DeAngelis, J. P., Ramappa, A. J., & Nazarian, A. (2023). A Validated Three-Dimensional, Heterogenous Finite Element Model of the Rotator Cuff and The Effects of Collagen Orientation. *Annals of biomedical engineering*, 51(5), 1002–1013. DOI: <https://doi.org/10.1007/s10439-022-03114-9>
- Williamson, P. M., Yeritsyan, D., Peacock, T., Chainani, P., Momenzadeh, K., Asciutto, D., Pathirana, P., Avakian, C., **Stewart, I.**, Kheir, N., Abbasian, M., DeAngelis, J. P., Ramappa, A. J., & Nazarian, A. (2023). A passive ankle dorsiflexion testing system to assess mechanobiological and structural response to cyclic loading in rat Achilles tendon. *Journal of biomechanics*, 156, 111664. DOI: [10.1016/j.jbiomech.2023.111664](https://doi.org/10.1016/j.jbiomech.2023.111664)
- Garcia, M., Caro, D., Buzo, M., **Stewart, I.**, Rodriguez D., King E., Momenzadeh, K., Kheir, Nadim., Khak, M., Abbasian M., DeAngelis Joseph, Ramappa, A., Nazarian, Ara. Effect of Glenohumeral Abduction on Supraspinatus Tendon

Strain: A Cadaveric Study. *SurgiColl.* 2024;2(3). DOI: [doi:10.58616/001c.121853](https://doi.org/10.58616/001c.121853)

- **Stewart, I.** (2023) Atomistic Modeling of Osseointegration: Binding affinities and modes for therapeutic proteins on Inorganic Surfaces. *Wellesley College thesis repository* (under embargo 2 years);

POSTERS

- **Stewart, I.**, Buzo Mena, M., Garcia, M., & Nazarian, A. (2023, February). *Influence of mouse femur positioning and orientation in 3-point bending framework: A finite element study* [**Poster presentation**]. Orthopaedic Research Society Annual Meeting (ORS 2023), Dallas, TX, United States.
- **Stewart, I.**, Buzo Mena, M., Garcia, M., & Nazarian, A. (2023, October). *High variabilities in mouse femur strength due to specimen malalignments and testing protocols* [**Poster presentation**]. American Society for Bone and Mineral Research Annual Meeting (ASBMR 2023), Vancouver, BC, Canada.